

Working with Data

- Strings:
 - String Indexing
 - String Methods
 - Format Specifiers
- Slicing and Concatenation

Strings

- Is a sequence of characters
- Enclosed in quotes (single, double, or triple quotes)
- Used to represent text in Python

String Indexing

- Access individual characters of a string by their index
- Indexing operator []
- [start : end : step]

Exercises

- Extract and print the last four digits of credit card numbers.

String Methods

Built-in functions that operate on strings

- `x.find("s")`
- `x.rfind("b")`
- `x.capitalize()`
- `x.title()`
- `x.upper()`
- `x.lower()`
- `x.isdigit()`
- `x.isalpha()`
- `x.count("-")`
- `x.replace("-", " ")`
- `x.split()`

Exercises

- Validate user input exercise
 - 1. Username is no more than 12 characters
 - 2. Username must not contain spaces
 - 3. Username must not contain digits

Format Specifiers

- Format a value based on what flags are inserted
- **{value : flags}**
- . (number)f = round decimal places
- : (number) = allocate spaces
- :03 = zero pad spaces
- :< = left justify
- :> = right justify
- :^ = center align
- :+ = use a plus sign
- : , = comma separator

Exercises

- Create a product price table displaying the price, tax (8%), and total price for three products, formatted with proper alignment and precision using f-strings.

Slicing and Concatenation

- Slicing: Extracts a part of a string using a start and end index
- Concatenation: Combines two or more strings into one
- Example: "Hello" + " World" results in "Hello World"

Exercises

- Extract username from email address